

Auditing crop host supervision in Swin models for condition recognition across multiple crops: evidence from 90 controlled runs

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Abstract

Classifiers covering multiple crops can return a condition that conflicts with their own crop prediction. A separate host head reveals these conflicts, although the host is already determined by the joint label. We tested whether host supervision improves recognition or mainly supports model audit. A Swin Small model pretrained on ImageNet was evaluated on three fixed benchmarks containing 43, 88, and 215 classes. Related augmentation families were confined to one split, exact duplicates and label conflicts were checked, and flat and dual models were paired by seed. Ten pairs were run per dataset, while two agreement losses were evaluated with five matched seeds, giving 90 runs. Flat and dual accuracies were 99.409 versus 99.398 percent, 97.657 versus 97.708 percent, and 92.151 versus 92.271 percent. At a predeclared margin of ± 0.25 percentage points, all outcomes on the benchmark with 43 classes were equivalent. Accuracy and cross-host error were also equivalent on the 88-class benchmark; macro F1 was unresolved. Neither agreement loss improved performance consistently across datasets and metrics. On

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the benchmark with 215 classes, a labeled-host oracle increased accuracy by about 2.2 points. Predicted-host masking removed all head conflicts but changed mean accuracy by at most 0.018 points. Temperature scaling fitted on validation data reduced expected calibration error from about 5.4 percent to between 1.6 and 1.7 percent. Host supervision did not yield a dependable accuracy gain; its clearest value was exposing internal disagreement for audit.

Keywords: Plant disease recognition, hierarchical classification, Swin Transformer, model audit, split integrity, probability calibration

1. Introduction

Image recognition is now used to organize crop image collections, flag likely disease cases, and support specialist review [2, 5]. A system that covers several crops must do more than choose a condition label. Its crop and condition outputs should agree, and the reported confidence should be useful when a prediction is accepted or sent for review. For example, a model may assign a tomato condition while its crop head selects grape. The condition may still be correct, but the model has produced two incompatible descriptions of the same image.

Many plant image datasets encode crop and condition in one joint class, such as `Tomato__late_blight`. A flat classifier treats these classes as an ordinary list. A second head can predict the crop host from the shared image features and can identify conflicts across the taxonomy. The extra target, however, is not independent in the datasets examined here. Once the joint class is known, the crop host follows from a fixed mapping. The host task repeats part of the original label, so an improvement cannot be assumed. A related concern applies to masking. Restricting condition classes to the crop predicted by the host head always produces a compatible pair, but an incorrect host prediction can remove the correct condition from consideration.

Small changes in benchmark accuracy are also easy to misread. Augmented copies of one photograph can enter different splits, exact duplicates can carry conflicting labels, and a favorable random seed can make one model appear better than another. Model rankings may change with initialization, data order, and other sources of training variation [18]. A sound comparison needs fixed split identities, paired runs, uncertainty for the paired difference, and a stated threshold for an effect that is too small to matter in practice. A nonsignificant test alone does not establish that two models perform similarly.

We asked whether deterministic crop host supervision adds useful evidence to a Swin model that predicts joint crop conditions. We addressed five questions. First, we compared condition accuracy, macro F1, and cross-host error between flat and dual models. Second, we tested practical equivalence at a margin declared before analysis and checked two alternative margins. Third, we evaluated MAE and $\text{KL}(q\|p^h)$ agreement losses. Fourth, we measured disagreement between the two heads before and after predicted-host masking, and compared the masking result with a labeled-host oracle that is unavailable at deployment. Finally, we examined calibration and selective error using temperatures fitted only on validation data.

The contribution lies in the controlled evaluation rather than in a new image backbone. Each flat and dual comparison used the same Swin Small representation, optimizer, split, and seeds. The three taxonomies differ markedly in size and imbalance, which lets us examine when the host head becomes more active and whether that activity improves the primary prediction. All conclusions refer to the fixed datasets used here. They do not establish performance on new farms, cameras, seasons, or geographic regions.

2. Related work

2.1. Recognition of crop conditions from images

PlantVillage made controlled evaluation of plant disease classifiers easier to reproduce [1, 2]. Convolutional networks later reached high accuracy on curated leaf images [3], while subsequent work showed that dataset size, background, visual diversity, and acquisition conditions strongly affect transfer beyond the source collection [4, 5]. Vision Transformers and shifted-window Transformers provide strong pretrained representations for image recognition [6, 7]. An improved Vision Transformer has also been tested for crop pest recognition in this journal [8]. We chose Swin Small as a common pretrained backbone so that the comparison focuses on the host target and not on a new feature extractor.

2.2. Hierarchical labels and auxiliary tasks

Multitask learning can help when an auxiliary target contributes information that supports the main task [11]. Hierarchical classification uses known relations among labels to organize predictions or prevent invalid outputs [12]. Previous studies have used crop identity as metadata, an intermediate prediction, or an additional target in conditioned networks and models with

multiple outputs [13, 14]. Our setting is narrower. Every host target can be recovered exactly from the joint condition label. The experiment therefore asks whether repeating that information changes the learned decision under otherwise matched training. It also separates an accurate prediction from one made valid by a rule after the model has produced conflicting outputs.

2.3. Evidence needed for a model audit

Agricultural image studies increasingly report more than overall accuracy. Recent reviews in this journal note that data composition and evaluation choices shape the conclusions drawn from plant health models [9, 10]. Ergün and Okumuş studied an adaptive ensemble on imbalanced eggplant data [15]. Singh et al. combined deduplication, calibration, and agronomic rule checks for rice disease diagnosis [16]. Outside agriculture, identity leakage has been linked to optimistic estimates in machine learning research [17].

This leaves a practical question. When part of a hierarchy is deterministic, does the auxiliary level improve the primary decision, or does it simply provide another view of the same output? We combine source identity controls, paired replication, equivalence tests, head disagreement rate (HDR), taxonomic consistency inference (TCI), a labeled-host oracle, and calibration. Table 1 places these elements beside selected studies that condition on crop identity, use several outputs, or apply rule checks.

3. Materials and methods

3.1. Study design

The three datasets were analyzed separately. No image moved from one benchmark to another. Split manifests and ordered mappings from conditions to hosts were fixed before model training. Each primary comparison paired a flat model and a dual model with the same seed. The two agreement losses were secondary analyses with fixed coefficients; their settings were not selected from test results. Figure 1 shows the data flow, Swin Small backbone, prediction heads, training losses, and locked test analyses.

3.2. Datasets and split integrity

Each image had one joint crop and condition label y . A fixed mapping $\phi(y)$ assigned that class to a host group. We use *condition* rather than *disease* because the label sets also include healthy states, pests, nutrient disorders, and one supervised non-plant category.

Table 1: Positioning against representative crop-condition studies. “NR” means that the cited article did not report the item as an explicit analysis; it does not establish that the underlying data or code lacked the property.

Study	Data setting	Hierarchy design	Identity control	Paired seeds	Equivalence / calibration	Constraint audit
Mohanty et al. [2]	Controlled PlantVillage images	Joint crop and condition CNN	NR	NR	NR / NR	NR
Picon et al. [13]	Multi-crop field images	Crop-metadata-conditioned CNN	NR	NR	NR / NR	NR
Yao et al. [14]	Three benchmark datasets	Multi-output and stacked CNNs	NR	NR	NR / NR	NR
Fu et al. [8]	Crop-pest image benchmark	Improved ViT classifier	NR	NR	NR / NR	NR
Ergün and Okumuş [15]	Three eggplant benchmarks	Learned CNN ensemble and TTA	NR	NR; CV	No / NR	NR
Singh et al. [16]	PaddyDoctor rice images	Post-hoc agronomic rule validator	Perceptual-hash deduplication	NR	No / temperature scaling	Rule contradictions
This study	Three fixed in-domain benchmarks	Auxiliary host head and masking	Original-family and/or exact-hash controls	10 paired	TOST, Holm, margin sensitivity / scaling	HDR, TCI, oracle host

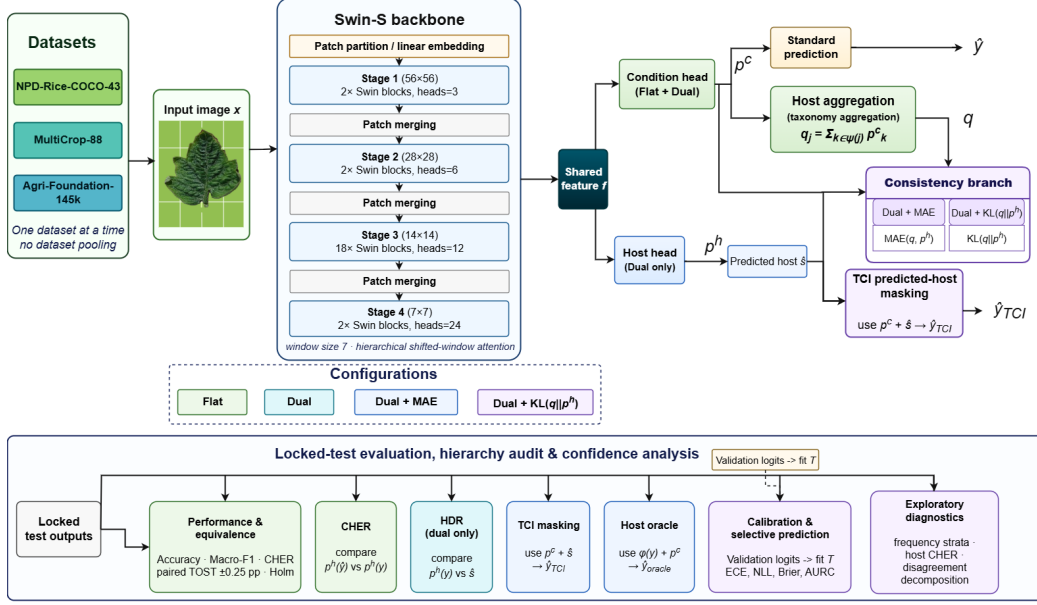


Figure 1: Study workflow and model architecture. A 224×224 image passes through the four stages of Swin Small, with stage depths 2, 2, 18, and 2 and attention head counts 3, 6, 12, and 24. Every configuration has a condition head. Dual configurations also have a host head. Condition probabilities are summed within the fixed taxonomy to form the condition-implied host distribution q . The MAE and $KL(q||p^h)$ branches add differentiable agreement losses during training. TCI restricts the taxonomy decision to the host selected by the direct host head. Locked test outputs are used for performance, equivalence, CHER, HDR, TCI, the labeled-host oracle, calibration, selective risk, and analyses by class frequency. CHER is the cross-host error rate; HDR is the head disagreement rate; TOST refers to two one-sided tests; ECE is expected calibration error; NLL is negative log likelihood; and AURC is the area under the risk-coverage curve. The workflow schematic was prepared with assistance from an LLM and was subsequently checked and edited by the authors.

NPD-Rice-COCO-43. The first benchmark was assembled locally from three public sources. The New Plant Diseases Dataset was the main component and contributed 87,425 local image paths in 38 crop-condition classes [25]. The Rice Leafs Dataset contributed all 3,355 local images in four classes: Brown Spot, Healthy, Hispa, and Leaf Blast [26].

To provide a supervised non-plant background category, a subset of 2,608 images was additionally selected from the COCO 2014 dataset [27, 28]. COCO was not used as an agricultural data source and did not contribute any plant-disease classes. Instead, the selected images represented unrelated visual content, including people, animals, and common everyday objects, and were grouped into a single non-plant background class. The purpose of this auxiliary category was to reduce the tendency of the classifier to assign clearly non-plant inputs to one of the crop-condition classes. Because these images were included during model training, this category is treated as a supervised background class rather than as an open-set or out-of-distribution evaluation. In the resulting NPD-Rice-COCO-43 benchmark, the COCO-derived category therefore constitutes one auxiliary non-plant class alongside the 42 crop-condition classes obtained from the New Plant Diseases Dataset and the Rice Leafs Dataset.

The complete benchmark contained 93,388 image paths, 43 prediction classes, and 16 taxonomy groups. The supplied archive does not retain the original COCO sampling script. We therefore make no unverified claim about category balance or a formal adversarial sampling rule for the selected background images.

The packaged source directories were pooled and split deterministically in an 80, 10, and 10 percent ratio. Augmented files linked by a source UUID or a normalized stem before augmentation were assigned together. Rice files with the same source stem and COCO files with the same image identifier were also kept together. Exact byte duplicates were joined to the same assignment group. No exact hash appeared with conflicting labels, and no path was removed.

MultiCrop-88. This local composite was derived from the Kaggle Plant Disease Classification Merged Dataset [30]. It initially contained 79,086 image paths in 88 classes and 23 host groups. The benchmark is also consistent with the multi-crop disease setting considered by Ahamed et al. [29]. We removed 1,254 redundant same-label paths from 1,232 exact-hash clusters. A further 62 paths from 29 hashes were excluded because the same bytes carried different

labels. The 77,770 remaining images were divided deterministically into 80, 10, and 10 percent partitions with every class present in each partition. Because the Kaggle package does not provide original acquisition or augmentation-family identifiers, `family_id` records an exact-hash assignment unit rather than an independently verified biological or acquisition family.

Agri-Foundation-145k. The Hugging Face distribution used locally and the permanent Zenodo record describe Agri-Foundation-145k as a 215-class agricultural disease dataset [31, 32]. The locked local derivative contained 144,752 paths in 215 classes and 36 host groups. It had 92,571 training images, 23,140 validation images, and a separate test set of 29,041 images. The one-path difference came from a synthetic 10×10 file named `dummy.jpg`. This file was the only development example for one joint class and is not an observed plant image. It was retained because the reported runs had already been completed with that locked derivative. The exact-hash audit found no match between the development pool and the test set. All 215 classes occurred in training and test, while 212 occurred in validation. Results for the class that contains the placeholder, and its contribution to absolute macro F1, require special caution. Since every configuration used the same derivative, the paired comparison remains case matched.

Table 2 reports the final counts and the identity checks. For NPD-Rice-COCO-43, no assignment group, source family, or SHA-256 hash crossed a split. For MultiCrop-88, no assignment group or exact hash crossed a split. Exact hashes do not identify near duplicates created by resizing, compression, cropping, or other image transformations. No perceptual hash or embedding similarity audit was available for MultiCrop-88 or Agri-Foundation-145k.

Table 2: Dataset counts and split integrity checks. Counts refer to paths after the stated exclusions. The local Agri-Foundation-145k total is one path above the public record because of the disclosed development placeholder.

Dataset	Hosts	Classes	Train	Validation	Test	Excluded	Integrity control
NPD-Rice-COCO-43	16	43	74,700	9,355	9,333	0	Source-family grouping; zero overlap by family or hash
MultiCrop-88	23	88	62,214	7,787	7,769	1,316	Exact deduplication; conflicting hashes excluded
Agri-Foundation-145k	36	215	92,571	23,140	29,041	0	Fixed split; one development placeholder; zero exact development-to-test overlap

3.3. Flat and host-aware Swin models

The backbone used the exact `timm` weight identifier `swin_small_patch4_window7_224.ms_in22k_ft_in1k`. These weights were obtained by pretraining on ImageNet 22k followed by fine-tuning on ImageNet 1k [33]. Swin Small

divides a 224×224 image into 4×4 patches, then applies four shifted-window stages with depths $(2, 2, 18, 2)$ and attention head counts $(3, 6, 12, 24)$. The final representation $\mathbf{f}$ has 768 dimensions [7]. The flat model has one linear condition head. The dual model adds a separate host head:

$$\mathbf{z}^c = W_c \mathbf{f} + b_c, \quad \mathbf{z}^h = W_h \mathbf{f} + b_h. \quad (1)$$

The added head contains $(768 + 1)S$ trainable parameters. This corresponds to 12,304 parameters for 16 hosts, 17,687 for 23 hosts, and 27,684 for 36 hosts. The backbone is identical in the two models.

Softmax gives condition probabilities $\mathbf{p}^c$ and direct host probabilities $\mathbf{p}^h$. Condition probabilities are also summed within the taxonomy to obtain a host distribution implied by the condition head:

$$q_j = \sum_{k:\phi(k)=j} p_k^c, \quad j = 1, \dots, S. \quad (2)$$

Standard inference selects $\hat{y} = \arg \max_k p_k^c$ and independently selects $\hat{s} = \arg \max_j p_j^h$. TCI was examined as a secondary analysis. It masks all condition classes outside the directly predicted host:

$$\hat{y}_{\text{TCI}} = \arg \max_{k:\phi(k)=\hat{s}} p_k^c. \quad (3)$$

This rule needs no retraining and always returns a condition that belongs to the reported host. It cannot guarantee that either prediction is correct.

The flat loss was condition cross entropy. The dual loss added host cross entropy with a fixed weight of one:

$$\mathcal{L}_{\text{dual}} = \text{CE}(\mathbf{z}^c, y) + \alpha \text{CE}(\mathbf{z}^h, \phi(y)), \quad \alpha = 1. \quad (4)$$

Two additional variants encouraged agreement between $\mathbf{q}$ and $\mathbf{p}^h$:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{S} \sum_{j=1}^S |q_j - p_j^h|, \quad (5)$$

$$\mathcal{L}_{\text{KL}(q\|p^h)} = \sum_{j=1}^S q_j \left[\log \{ \max(q_j, \epsilon) \} - \log \{ \max(p_j^h, \epsilon) \} \right]. \quad (6)$$

The full loss was $\mathcal{L}_{\text{dual}} + \lambda \mathcal{L}_{\text{MAE}}$ or $\mathcal{L}_{\text{dual}} + \lambda \mathcal{L}_{\text{KL}(q\|p^h)}$, with $\lambda = 1$. MAE was averaged over samples and host categories. KL was summed over

host categories and averaged over samples. The numerical floor was $\epsilon = \text{torch.finfo}(\mathbf{p}^h.\text{dtype}).\text{eps}$. The condition-implied distribution was not detached, so both heads received gradients from the agreement term. The unit coefficients were fixed and do not equalize the gradient scales of MAE and KL.

3.4. Training protocol and experiment matrix

Every run started from the same pretrained Swin Small weights. We used AdamW [19], a learning rate of 10^{-4} , weight decay of 10^{-4} , batch size 16, cosine annealing without warmup, and 30 epochs. The training transform used a random resized crop to 224×224 pixels with a scale from 0.5 to 1.0, a random horizontal flip, and color jitter of 0.2 for brightness, contrast, and saturation. Validation and test images were resized to 256 pixels, center-cropped to 224×224 , and normalized with ImageNet statistics. The checkpoint with the highest validation macro F1 was retained.

Training used CUDA automatic mixed precision. Selected checkpoint weights were stored in FP32. The primary runs exported FP32 validation and test logits for calibration. Saved predictions were used for the discrete hierarchy analyses, and logits were used when available. Seeds controlled initialization, data order, and augmentation. Deterministic PyTorch and cuDNN behavior was enabled, `CUBLAS_WORKSPACE_CONFIG=:4096:8` was set, and random number states were stored for resumed runs.

Flat and dual models used seeds 2026 to 2035 on each dataset. The MAE and KL variants used seeds 2026 to 2030 and were compared with dual controls using the same five seeds. This produced 30 runs for each dataset and 90 runs in total (Table 3). Every retained run completed all 30 epochs.

Table 3: Experiment matrix. Each seed represents a separate fine-tuning run.

Configuration	Objective	Seeds per dataset	Runs
Flat	Condition CE	10	30
Dual	Condition CE plus host CE	10	30
Dual plus MAE	Dual loss plus MAE agreement	5	15
Dual plus KL($q\ p^h$)	Dual loss plus KL agreement	5	15
Total			90

3.5. Outcomes and hierarchy diagnostics

The primary outcomes were top-1 condition accuracy, macro F1, and cross-host error rate (CHER). Macro F1 was the unweighted mean over all condition classes; a class received zero when its F1 value was undefined. CHER counts a condition prediction as a cross-host error when it belongs to a different host from the label:

$$\text{CHER} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\phi(\hat{y}_i) \neq \phi(y_i)]. \quad (7)$$

For dual models, HDR compares the host implied by the condition argmax with the host selected by the direct host head:

$$\text{HDR} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\phi(\hat{y}_i) \neq \hat{s}_i]. \quad (8)$$

HDR measures internal agreement. CHER compares the condition prediction with the labeled host. Host accuracy and HDR were calculated for all 60 dual runs. TCI was then applied to the same locked outputs, and we recorded its change in accuracy, macro F1, and CHER. HDR and TCI are not defined for the flat model. Unless noted otherwise, metrics are reported as percentages.

We ran additional descriptive analyses on the benchmark with 215 classes. Classes were grouped by training support: 1 to 25, 26 to 100, 101 to 500, and more than 500 images. Class F1 was averaged within each group for every seed. CHER was also summarized by labeled host. A labeled-host oracle masked fixed condition logits to classes compatible with the true host before taking the argmax. The oracle sets CHER to zero and is not deployable; it estimates how much error lies across host boundaries. For the dual model without an agreement loss, each head disagreement was classified according to whether the direct host, the condition-implied host, or neither matched the label. We also counted correct predictions gained and lost after TCI. These analyses were exploratory and were not included in the adjusted superiority tests.

3.6. Calibration and statistical analysis

Within each dataset, flat and dual outcomes were paired by seed. We calculated the mean dual minus flat difference and a 95 percent confidence interval based on the paired t distribution. Before examining the differences,

we defined practical equivalence as an effect inside ± 0.25 percentage points. For accuracy, this is one changed decision in 400 images. We applied the same absolute margin to macro F1 and CHER so that the threshold was not widened after seeing the results.

Equivalence was tested with two one-sided tests using a 90 percent interval. The nine decisions formed by three datasets and three metrics were adjusted by the Holm procedure [23, 24]. We declared equivalence only when the entire 90 percent interval lay within the margin and the adjusted test passed at $\alpha = 0.05$. An outcome that did not pass was called unresolved; it was not treated as evidence that the models differ. The same nine-decision adjustment was repeated after analysis at margins of ± 0.10 and ± 0.50 points. We also translated each CHER margin into a number of errors on the fixed test set. These intervals describe variation across fine-tuning seeds for one test set. They do not represent repeated sampling from a new population.

NPD-Rice-COCO-43 contains different numbers of augmented files per source family. As a sensitivity check, every test family received total weight one. Each image was weighted by the inverse of its family size, and weighted accuracy, macro F1, and CHER were calculated for each seed before the paired summary. This changed the contribution of repeated augmentations without changing the models or test cases.

The five-seed comparisons of agreement losses were secondary. We report paired 95 percent intervals but do not use them for a general claim of superiority.

Probability quality was evaluated for all 60 primary flat and dual checkpoints. A single temperature was fitted to validation logits by minimizing negative log likelihood (NLL) and then applied once to the locked test logits [20]. We report expected calibration error (ECE) from 15 top-label bins, NLL, the summed multiclass Brier score [21], and area under the risk-coverage curve (AURC) [22]. Lower values are better. Risk-coverage curves rank images by maximum calibrated condition probability and calculate the error among the retained highest-confidence fraction. Temperature scaling changes probability magnitude but normally leaves the predicted class and its ordering by confidence unchanged.

3.7. LLM support in the research workflow

An assistant based on a large language model (LLM) supported code review, internal consistency checks, manuscript organization, language editing, and preparation of the workflow schematic. The LLM did not generate

training images, experimental measurements, model predictions, or statistical decisions. The authors inspected the implementation, checked the references, and matched every reported value to the stored outputs. They take responsibility for the final methodology, analysis, and presentation.

4. Results

4.1. Integrity checks and completed runs

The reconstructed NPD-Rice-COCO-43 and MultiCrop-88 partitions passed all specified exact identity checks. NPD-Rice-COCO-43 had no overlap across partitions by assignment group, source family, or SHA-256 hash. MultiCrop-88 retained 77,770 unique images after 1,316 exclusions and had no shared assignment group or hash across partitions. Its hash-derived `family_id` should not be interpreted as independent source provenance. The Agri-Foundation-145k test set had no exact hash match with the development pool. The development placeholder described in Section 3.2 remained in the locked local derivative.

All 90 runs completed 30 epochs and passed the available checks of configuration, identity, and output files. Each run recorded the intended dataset, seed, model configuration, checkpoint history, and locked test predictions. FP32 validation and test logits were available for every primary flat and dual run.

4.2. Flat and dual performance

Table 4 gives the main test results. NPD-Rice-COCO-43 was close to saturation. The dual model changed accuracy by -0.011 points and macro F1 by -0.054 points relative to the flat model. On MultiCrop-88, the dual model was 0.050 points higher in accuracy and 0.194 points higher in macro F1, with CHER lower by 0.030 points. On Agri-Foundation-145k, accuracy was 0.121 points higher, macro F1 was 0.571 points higher, and CHER was 0.183 points lower.

The paired equivalence analysis places these differences in context (Table 5; Fig. 2). All three NPD-Rice-COCO-43 outcomes met the prespecified criterion. MultiCrop-88 accuracy and CHER also met it. Its macro F1 interval crossed the upper margin and remained unresolved. None of the three Agri-Foundation-145k outcomes passed the primary equivalence test. For CHER, the 95 percent interval was $[-0.270, -0.096]$ points, which favors the dual model, but the 90 percent TOST interval extended to -0.253 , just

Table 4: Fixed-protocol test performance over paired seeds. Values are percentages reported as mean $\pm$ standard deviation.

Dataset	Configuration	Seeds	Accuracy	Macro-F1	CHER
NPD-Rice-COCO-43	Flat	10	99.409 $\pm$ 0.069	98.538 $\pm$ 0.120	0.038 $\pm$ 0.018
NPD-Rice-COCO-43	Dual, no penalty	10	99.398 $\pm$ 0.031	98.483 $\pm$ 0.063	0.024 $\pm$ 0.005
MultiCrop-88	Flat	10	97.657 $\pm$ 0.136	95.138 $\pm$ 0.310	0.336 $\pm$ 0.040
MultiCrop-88	Dual, no penalty	10	97.708 $\pm$ 0.098	95.333 $\pm$ 0.319	0.306 $\pm$ 0.043
Agri-Foundation-145k	Flat	10	92.151 $\pm$ 0.164	62.645 $\pm$ 0.640	3.321 $\pm$ 0.097
Agri-Foundation-145k	Dual, no penalty	10	92.271 $\pm$ 0.118	63.216 $\pm$ 0.791	3.138 $\pm$ 0.076

Table 5: Paired equivalence analysis for dual-head minus flat performance. All differences and intervals are percentage points; the predeclared margin is ± 0.25 percentage points.

Dataset	Metric	Mean difference	95% CI	90% TOST CI	Holm decision
NPD-Rice-COCO-43	Accuracy	-0.011	[-0.065, 0.043]	[-0.054, 0.033]	Equivalent ($p_{\text{Holm}} = 1.20 \times 10^{-5}$)
NPD-Rice-COCO-43	Macro-F1	-0.054	[-0.151, 0.043]	[-0.133, 0.025]	Equivalent ($p_{\text{Holm}} = 0.00413$)
NPD-Rice-COCO-43	CHER	-0.014	[-0.028, 0.000]	[-0.025, -0.003]	Equivalent ($p_{\text{Holm}} = 1.32 \times 10^{-10}$)
MultiCrop-88	Accuracy	0.050	[-0.049, 0.150]	[-0.030, 0.131]	Equivalent ($p_{\text{Holm}} = 0.00413$)
MultiCrop-88	Macro-F1	0.194	[-0.120, 0.509]	[-0.061, 0.450]	Not established ($p_{\text{Holm}} = 0.699$)
MultiCrop-88	CHER	-0.030	[-0.063, 0.003]	[-0.056, -0.003]	Equivalent ($p_{\text{Holm}} = 4.32 \times 10^{-7}$)
Agri-Foundation-145k	Accuracy	0.121	[-0.018, 0.259]	[0.009, 0.232]	Not established ($p_{\text{Holm}} = 0.126$)
Agri-Foundation-145k	Macro-F1	0.571	[-0.123, 1.265]	[0.009, 1.134]	Not established ($p_{\text{Holm}} = 0.839$)
Agri-Foundation-145k	CHER	-0.183	[-0.270, -0.096]	[-0.253, -0.112]	Not established ($p_{\text{Holm}} = 0.172$)

outside the -0.25 boundary. This result is compatible with a small benefit, but it does not establish equivalence or a general performance advantage.

The margin analysis showed how the decision changed with the chosen threshold (Table 6). With a ± 0.10 point margin, NPD-Rice-COCO-43 accuracy and CHER and MultiCrop-88 CHER remained equivalent. With a ± 0.50 point margin, accuracy and CHER on Agri-Foundation-145k also became equivalent. Macro F1 remained unresolved on the two larger taxonomies. The interpretation of the hardest benchmark therefore depends partly on the size of an effect judged negligible.

CHER counts make the margins easier to interpret. The mean flat and dual counts were 3.5 and 2.2 errors among 9,333 NPD-Rice-COCO-43 test images, 26.1 and 23.8 among 7,769 MultiCrop-88 images, and 964.5 and 911.4 among 29,041 Agri-Foundation-145k images. A 0.10 point margin equals 9.3, 7.8, and 29.0 events in these test sets. The primary 0.25 point margin equals 23.3, 19.4, and 72.6 events.

Giving each of the 5,171 NPD-Rice-COCO-43 test families equal total weight did not alter the comparison. Flat and dual means were 99.033 and 99.020 percent for accuracy, 98.522 and 98.491 percent for macro F1, and

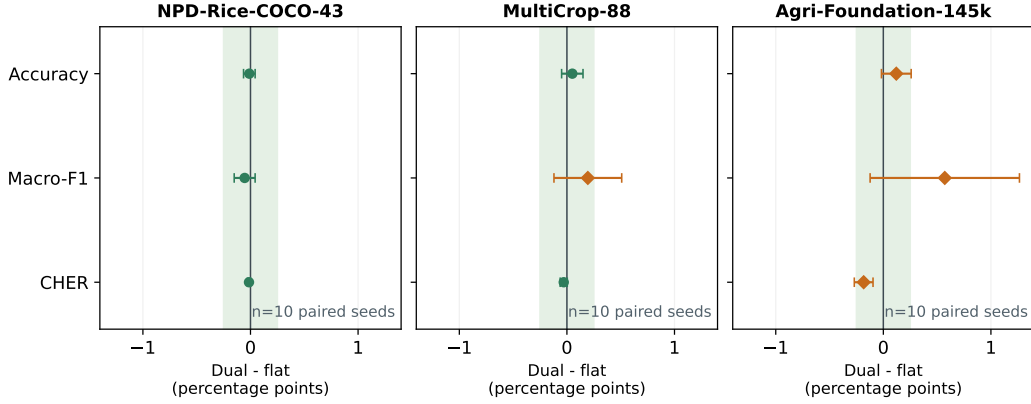


Figure 2: Paired effects for the dual model minus the flat model with 95 percent confidence intervals. Each dataset has 10 paired seeds. The shaded area marks the prespecified equivalence region of ± 0.25 points. Circles identify Holm-adjusted equivalence decisions; diamonds identify unresolved decisions. A negative ChER difference favors the dual model.

Table 6: Sensitivity of the Holm-adjusted equivalence decisions to the absolute margin. Each column applies Holm adjustment across the same nine dataset and metric decisions. The ± 0.25 -point column is the predeclared primary analysis; the other columns are post-hoc sensitivity checks.

Dataset	Metric	± 0.10 points	± 0.25 points	± 0.50 points
NPD-Rice-COCO-43	Accuracy	Equivalent	Equivalent	Equivalent
NPD-Rice-COCO-43	Macro-F1	Not established	Equivalent	Equivalent
NPD-Rice-COCO-43	CHER	Equivalent	Equivalent	Equivalent
MultiCrop-88	Accuracy	Not established	Equivalent	Equivalent
MultiCrop-88	Macro-F1	Not established	Not established	Not established
MultiCrop-88	CHER	Equivalent	Equivalent	Equivalent
Agri-Foundation-145k	Accuracy	Not established	Not established	Equivalent
Agri-Foundation-145k	Macro-F1	Not established	Not established	Not established
Agri-Foundation-145k	CHER	Not established	Not established	Equivalent

Table 7: Secondary penalty ablation at consistency weight $\lambda = 1$. All metrics are percentages reported as mean $\pm$ standard deviation over five matched seeds.

Dataset	Configuration	Seeds	Accuracy	Macro-F1	CHER
NPD-Rice-COCO-43	Dual, no penalty	5	99.404 ± 0.043	98.507 ± 0.060	0.021 ± 0.000
NPD-Rice-COCO-43	Dual + MAE	5	99.389 ± 0.039	98.451 ± 0.116	0.024 ± 0.005
NPD-Rice-COCO-43	Dual + $\text{KL}(q\ p^h)$	5	99.366 ± 0.044	98.438 ± 0.103	0.026 ± 0.010
MultiCrop-88	Dual, no penalty	5	97.657 ± 0.079	95.156 ± 0.329	0.304 ± 0.034
MultiCrop-88	Dual + MAE	5	97.732 ± 0.149	95.491 ± 0.238	0.283 ± 0.035
MultiCrop-88	Dual + $\text{KL}(q\ p^h)$	5	97.704 ± 0.150	95.107 ± 0.514	0.275 ± 0.023
Agri-Foundation-145k	Dual, no penalty	5	92.279 ± 0.092	63.139 ± 0.548	3.107 ± 0.056
Agri-Foundation-145k	Dual + MAE	5	92.124 ± 0.152	62.689 ± 0.659	3.208 ± 0.084
Agri-Foundation-145k	Dual + $\text{KL}(q\ p^h)$	5	92.388 ± 0.036	63.558 ± 0.205	3.034 ± 0.052

0.0216 and 0.0107 percent for CHER. The paired differences were -0.0129 points (95 percent CI $[-0.0829, 0.0571]$), -0.0307 points ($[-0.1352, 0.0739]$), and -0.0110 points ($[-0.0241, 0.0022]$). Large augmentation families were not responsible for the near-zero result.

4.3. Agreement losses

Neither agreement loss improved every dataset and metric (Table 7; Fig. 3). On NPD-Rice-COCO-43, MAE and KL changed macro F1 by -0.056 and -0.069 points compared with the matched dual control. On MultiCrop-88, MAE increased macro F1 by 0.335 points and lowered CHER by 0.021 points. KL changed macro F1 by -0.048 points and lowered CHER by 0.028 points. On Agri-Foundation-145k, KL produced the largest favorable secondary values: macro F1 rose by 0.419 points and CHER fell by 0.073 points. The intervals from five seeds were broad, especially for macro F1, and the directions were not stable across datasets.

4.4. Head disagreement and predicted-host masking

Head disagreement became more frequent as the taxonomy grew (Table 8). Mean HDR before TCI was no greater than 0.004 percent on NPD-Rice-COCO-43, ranged from 0.067 to 0.102 percent on MultiCrop-88, and ranged from 0.861 to 1.148 percent on Agri-Foundation-145k. TCI reduced HDR to zero in all 60 dual runs because the condition choice was restricted to the direct host prediction.

The forced agreement did not yield a stable accuracy gain. Across the nine combinations of dataset and dual configuration, mean accuracy changes ranged from -0.014 to $+0.018$ points. CHER decreased most on the benchmark with

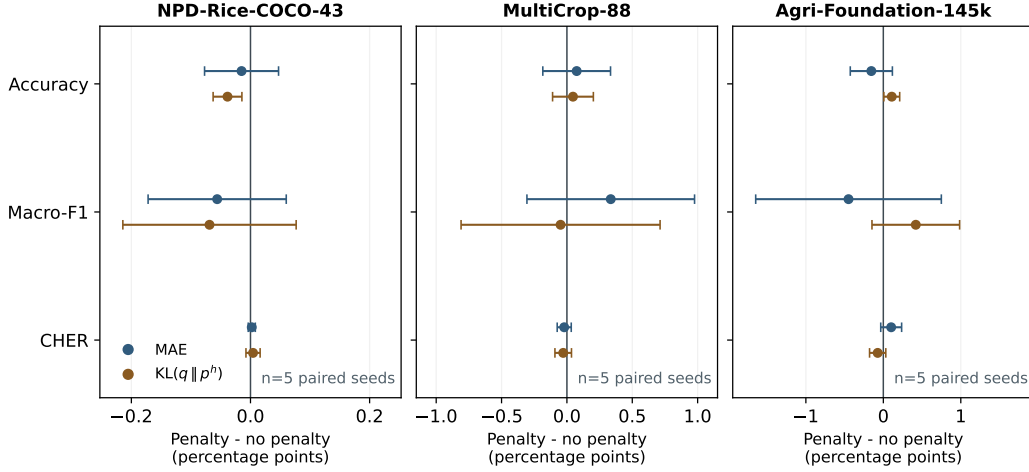


Figure 3: Paired effects of MAE and $KL(q||p^h)$ relative to the dual model without an agreement loss. Points show mean differences and bars show 95 percent confidence intervals from five paired seeds. Positive changes are favorable for accuracy and macro F1; negative changes are favorable for ChER.

215 classes: by 0.052 points for KL, 0.100 points for MAE, and 0.137 points for the dual model without an agreement loss. TCI removed some cross-host errors, but the small accuracy changes show that a compatible output is not necessarily a correct output.

For the dual model without a penalty on Agri-Foundation-145k, TCI changed an average of 332.3 predictions per seed, or 1.144 percent of the test set. Among these cases, only the direct host head matched the labeled host in 38.7 percent, only the condition-implied host matched it in 26.8 percent, and both were wrong in 34.5 percent. TCI gained 79.1 correct top-1 decisions per seed and lost 74.4, for a net gain of 4.7 cases, or 0.016 points. The small net effect came from helpful and harmful changes that nearly cancelled each other.

4.5. Class frequency and labeled-host oracle

The class-frequency analysis identified where the numerical dual effect was concentrated (Table 9; Fig. 4). Mean changes in macro F1 within the four training-support groups were 0.871 points for 1 to 25 images, 0.600 points for 26 to 100, 0.854 points for 101 to 500, and 0.035 points for more than 500. Only the 101 to 500 group had a paired 95 percent interval fully above zero ([0.533, 1.176]). These intervals were exploratory and were not adjusted for

Table 8: Head disagreement under standard inference and the effect of taxonomic consistency inference (TCI), implemented as predicted-host masking. Values are mean $\pm$ standard deviation over matched runs. Pre-TCI HDR is a percentage; deltas are TCI minus standard inference in percentage points. The full host-accuracy and macro-F1 results are provided in Supplementary Table S2.

Dataset	Configuration	Seeds	Pre-TCI HDR	Δ accuracy	Δ CHER
NPD-Rice-COCO-43	Dual, no penalty	10	0.004 ± 0.006	0.001 ± 0.003	-0.001 ± 0.003
NPD-Rice-COCO-43	Dual + MAE	5	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
NPD-Rice-COCO-43	Dual + $\text{KL}(q\ p^h)$	5	0.002 ± 0.005	0.002 ± 0.005	-0.002 ± 0.005
MultiCrop-88	Dual, no penalty	10	0.102 ± 0.026	0.013 ± 0.033	-0.022 ± 0.037
MultiCrop-88	Dual + MAE	5	0.072 ± 0.027	-0.005 ± 0.020	0.008 ± 0.020
MultiCrop-88	Dual + $\text{KL}(q\ p^h)$	5	0.067 ± 0.011	0.018 ± 0.030	-0.021 ± 0.023
Agri-Foundation-145k	Dual, no penalty	10	1.144 ± 0.052	0.016 ± 0.036	-0.137 ± 0.050
Agri-Foundation-145k	Dual + MAE	5	1.148 ± 0.044	-0.002 ± 0.039	-0.100 ± 0.044
Agri-Foundation-145k	Dual + $\text{KL}(q\ p^h)$	5	0.861 ± 0.047	-0.014 ± 0.046	-0.052 ± 0.031

multiple testing. The largest class changes occurred when test support was very small. `raspberry_spot`, for example, gained 23.19 F1 points with four test images, while `tomato_two_spotted_spider_mites` lost 20 points with one test image. Such values indicate instability and should not be read as reliable class rankings.

Table 9: Exploratory diagnostics for Agri-Foundation-145k. Panel A reports within-bin macro-F1 over 10 paired seeds, where bins are defined by training images per class. Panel B applies ground-truth-host oracle masking to fixed condition logits as a non-deployable oracle. Differences are dual minus flat in Panel A and oracle minus standard in Panel B. Panel A intervals are paired 95% confidence intervals across seeds; Panel B reports seed means.

Panel A: training-frequency strata

Training images/class	Classes	Training images	Flat macro-F1 (%)	Dual macro-F1 (%)	Difference (95% CI), points
1–25	42	610	30.071	30.942	0.871 [−1.677, 3.419]
26–100	81	4,181	52.528	53.128	0.600 [−0.174, 1.374]
101–500	42	7,344	75.327	76.181	0.854 [0.533, 1.176]
> 500	50	80,436	95.745	95.780	0.035 [−0.095, 0.164]

Panel B: ground-truth-host oracle

Configuration	Standard accuracy	Oracle accuracy	Δ accuracy	Standard macro-F1	Oracle macro-F1	Δ macro-F1
Flat	92.151	94.445	2.294	62.645	73.246	10.601
Dual, no penalty	92.271	94.456	2.184	63.216	73.448	10.232

Several hosts with modest test support accounted for much of the cross-host error. Flat and dual mean CHER values were 53.5 and 51.4 percent for

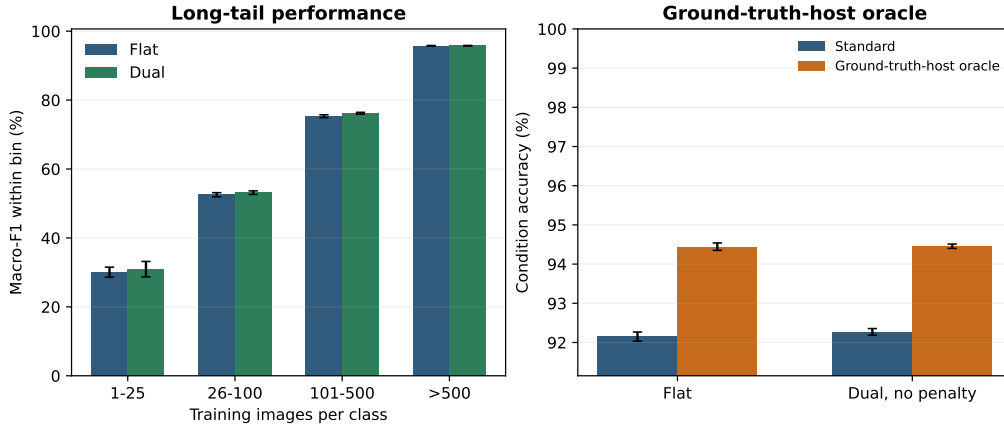


Figure 4: Class frequency and labeled-host oracle results for Agri-Foundation-145k. The left panel shows mean macro F1 within each training support group over 10 seeds; error bars are 95 percent confidence intervals for the seed mean. The right panel compares standard condition accuracy with accuracy after fixed condition logits are restricted by the labeled host. The oracle is not deployable and sets CHER to zero.

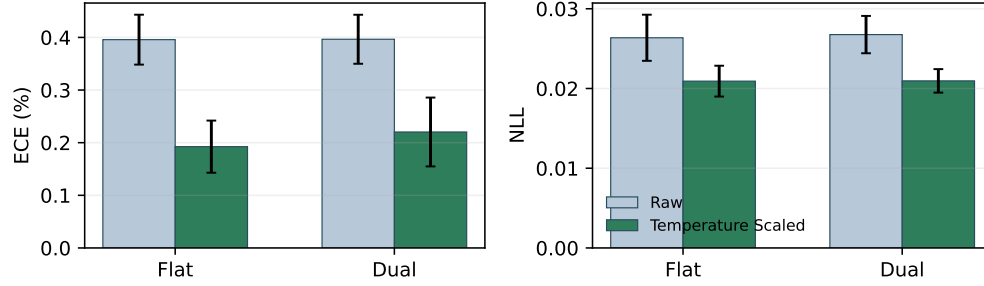
zucchini ($n = 100$), 49.18 and 47.84 percent for bean ($n = 134$), 45.65 and 44.26 percent for plum ($n = 115$), and 38.25 and 34.60 percent for tobacco ($n = 63$). The labeled-host oracle raised flat accuracy from 92.151 to 94.445 percent and dual accuracy from 92.271 to 94.456 percent. Macro F1 rose from 62.645 to 73.246 percent for the flat model and from 63.216 to 73.448 percent for the dual model. Accurate host information from outside the image classifier could remove a substantial group of errors. A host predicted from the same image did not provide the same benefit.

4.6. Probability quality and selective risk

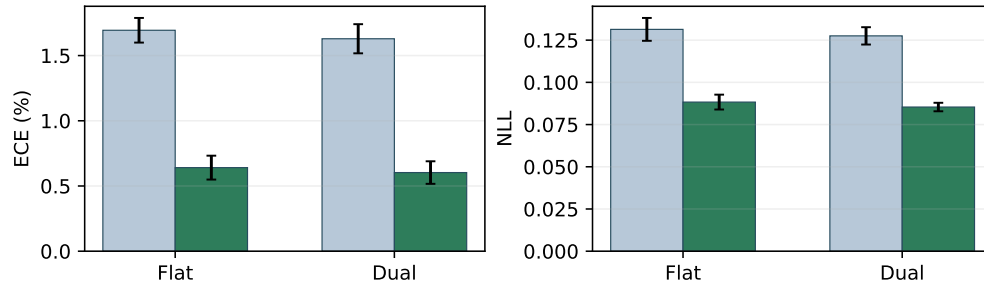
Raw calibration error increased with task difficulty (Table 10). ECE was about 0.396 percent on NPD-Rice-COCO-43, 1.629 to 1.694 percent on MultiCrop-88, and 5.378 to 5.446 percent on Agri-Foundation-145k. Temperature scaling reduced these ranges to 0.192 to 0.220 percent, 0.603 to 0.641 percent, and 1.560 to 1.701 percent. NLL and Brier score improved for every dataset and configuration.

AURC changed little because a single temperature does not materially reorder confidence scores. The flat and dual risk-coverage curves were also close within each dataset (Fig. 6). Referral based on confidence could reduce error among retained cases in this retrospective analysis. The curves do not

NPD-Rice-COCO-43



MultiCrop-88



Agri-Foundation-145k

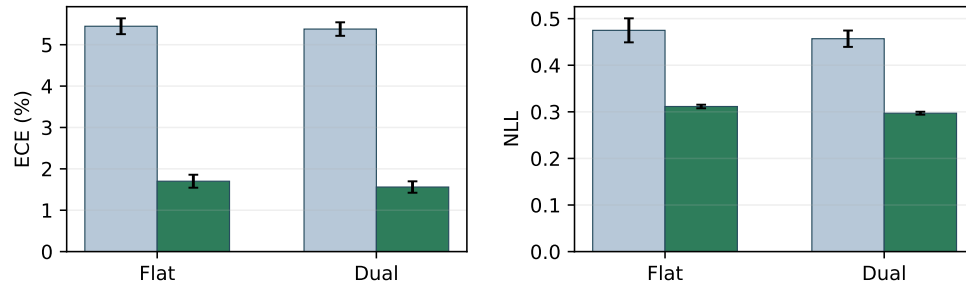


Figure 5: Expected calibration error and negative log likelihood before and after temperature scaling fitted on validation data. Bars show means over 10 seeds; error bars show standard deviations across seeds.

Table 10: Primary-checkpoint probability quality before and after validation-only temperature scaling. Arrows denote raw $\rightarrow$ temperature-scaled values; entries are means over 10 seeds.

Dataset	Configuration	Seeds	ECE (%)	NLL	Brier	AURC
NPD-Rice-COCO-43	Flat	10	0.396 $\rightarrow$ 0.192	0.0264 $\rightarrow$ 0.0209	0.0100 $\rightarrow$ 0.0096	0.0002 $\rightarrow$ 0.0002
NPD-Rice-COCO-43	Dual, no penalty	10	0.396 $\rightarrow$ 0.220	0.0268 $\rightarrow$ 0.0210	0.0101 $\rightarrow$ 0.0095	0.0002 $\rightarrow$ 0.0002
MultiCrop-88	Flat	10	1.694 $\rightarrow$ 0.641	0.1314 $\rightarrow$ 0.0883	0.0403 $\rightarrow$ 0.0371	0.0014 $\rightarrow$ 0.0016
MultiCrop-88	Dual, no penalty	10	1.629 $\rightarrow$ 0.603	0.1275 $\rightarrow$ 0.0854	0.0388 $\rightarrow$ 0.0360	0.0013 $\rightarrow$ 0.0014
Agri-Foundation-145k	Flat	10	5.446 $\rightarrow$ 1.701	0.4749 $\rightarrow$ 0.3114	0.1300 $\rightarrow$ 0.1139	0.0072 $\rightarrow$ 0.0073
Agri-Foundation-145k	Dual, no penalty	10	5.378 $\rightarrow$ 1.560	0.4569 $\rightarrow$ 0.2971	0.1278 $\rightarrow$ 0.1114	0.0071 $\rightarrow$ 0.0071

include reviewer capacity, review cost, or the accuracy of the person who receives a referred image.

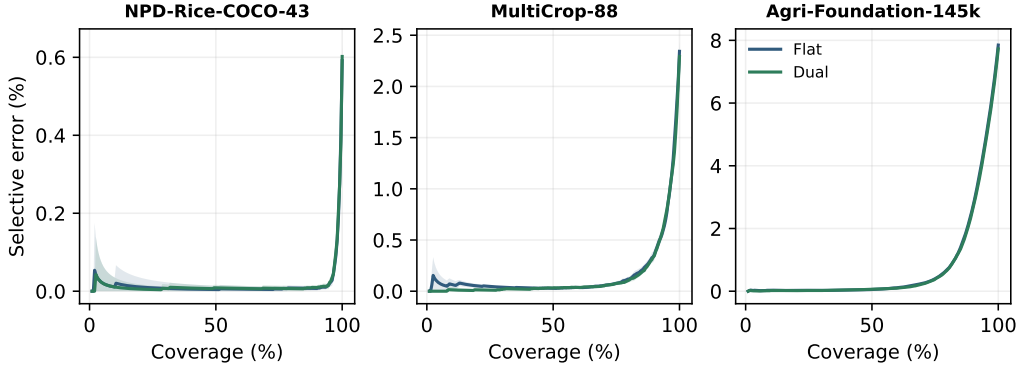


Figure 6: Selective condition error after temperature scaling as retained coverage changes. Lines show seed means for the primary flat and dual models; shaded bands show 95 percent confidence intervals across seeds. Lower error is better.

5. Discussion

5.1. What the host head changed

The host head produced no stable improvement in condition recognition. NPD-Rice-COCO-43 showed practical equivalence for all three primary outcomes. MultiCrop-88 showed equivalence for accuracy and CHER, while macro F1 remained unresolved. The largest numerical changes appeared on Agri-Foundation-145k, but the uncertainty was too wide to support either equivalence at the primary margin or a broad claim that the dual model was better. The gap of nearly 29 points between accuracy and macro F1 on this

dataset also confirms that overall accuracy alone is inadequate for a large imbalanced taxonomy.

The frequency and oracle analyses explain why host information can still matter. The clearest descriptive increase for the dual model occurred among classes with 101 to 500 training images. Results for the rarest classes varied widely because some test classes contained only one to four images. When the true host was supplied as an oracle, both models gained about 2.2 accuracy points and more than 10 macro F1 points. Reliable crop metadata can thus be useful. That benefit cannot be attributed to a host head trained from the same joint label and the same image features.

5.2. Agreement and correctness are different questions

HDR records conflicts before a rule is applied. TCI removes those conflicts by forcing the condition decision into the host selected by the direct head. An HDR after TCI of zero is guaranteed, so it says nothing by itself about whether the condition is correct. On Agri-Foundation-145k, TCI changed hundreds of predictions per seed, yet the gains and losses almost balanced. Reporting only the final agreement would hide this exchange. A useful audit should include HDR before masking together with the changes in accuracy, macro F1, and CHER.

The MAE and KL results lead to the same interpretation. Both losses encourage two probability distributions to agree, but both distributions ultimately come from targets contained in the same joint label. Once condition and host cross entropy have largely aligned the heads, another agreement term can change the optimization path without adding new information. The mixed effects from five seeds, along with different gradient scales under a common unit coefficient, do not support either loss as a generally useful addition.

5.3. Identity controls and uncertainty

The data checks define the scope of the comparison. Related augmented files were kept together in NPD-Rice-COCO-43. MultiCrop-88 was cleaned for exact duplicates and conflicting hashes. The Agri-Foundation-145k test set had no exact match in the development pool. Family weighting also showed that large augmentation groups did not drive the NPD-Rice-COCO-43 result. Fixed manifests and paired seeds make each model comparison traceable.

These checks cannot rule out every visually related image. MultiCrop-88 lacks independent acquisition provenance, and exact hashing cannot find a

resized, cropped, or recompressed copy. The synthetic development placeholder in Agri-Foundation-145k creates a separate problem because one class has no real development image. Since the same derivative was used for all configurations, the paired architecture comparison is still matched. Absolute macro F1 and the result for that class should not be generalized. The supplied class-exclusion script needs to be run and its output deposited before submission.

Equivalence testing also changed how the numerical differences should be read. A large p value would only show that a difference was not detected. The prespecified margin allowed a direct test of whether the whole plausible effect was small enough to ignore. The two sensitivity margins show which decisions remain stable and which depend on accepting a larger effect as negligible.

5.4. *Confidence and use in practice*

Temperature scaling was the only adjustment that improved all primary settings for ECE, NLL, and Brier score. This matters when probabilities are displayed to users or used for referral. The nearly unchanged AURC provides a useful limit: calibration made the probability values more faithful, but it did not make the model better at ranking easy and difficult images. It also did not correct wrong labels or demonstrate safety in field use.

These findings suggest a practical evaluation sequence. A host-aware model should first be compared with a flat baseline using the same seeds and a margin chosen before test results are examined. If the host head is retained for audit, disagreement should be reported before a taxonomy rule is applied. Any masking step should then be assessed by the correct decisions it gains and loses. A predicted host should not be described as independent crop evidence. When confidence will control acceptance or referral, calibration should be fitted on validation data and tested on untouched outputs.

Although the experiments used Swin Small, the same logic applies to other models whenever one level of a label hierarchy is determined by another. In such a setting, split identity, paired uncertainty, direct equivalence tests, unmasked head disagreement, and calibrated probabilities provide more information than a single best run.

6. Scope and limitations

This work evaluates performance within three fixed datasets. It does not test transfer across data sources, seasons, geographic regions, devices, or field

conditions. The three datasets are local derivatives with different class counts, host counts, imbalance, provenance, and visual properties. NPD-Rice-COCO-43 has source-family controls. MultiCrop-88 and Agri-Foundation-145k have exact-byte checks but no perceptual hash or embedding similarity analysis. The supervised COCO background class was represented during training and must not be described as open-set recognition or as general detection of inputs outside the training distribution.

The Agri-Foundation-145k derivative contains a synthetic 10×10 development placeholder as the only development support for one class. This limits the interpretation of absolute macro F1 and class-specific performance. No class-exclusion result is reported because the output from the available sensitivity script was absent from the supplied archive. That analysis should be run, checked, and included in the permanent repository. The present comparisons remain matched because all models were trained and tested on the same locked cases.

The intervals from 10 seeds quantify variation from fine-tuning on fixed test sets; they do not quantify uncertainty over new populations. Macro F1 intervals on the benchmark with 215 classes are broad. The host loss weight was fixed at $\alpha = 1$, and the MAE and KL analyses used five seeds with $\lambda = 1$. Other weights may behave differently. TCI uses a host predicted from the same representation, while the labeled-host oracle is not deployable. Analyses by frequency, class, host, and disagreement type were exploratory, and some of the largest class effects came from very small test supports.

Swin Small was pretrained on natural images. The findings concern fine-tuning this backbone and do not cover training from scratch or every possible architecture. Calibration and selective risk were evaluated retrospectively. No field workflow, specialist review process, latency constraint, user study, or agronomic outcome was tested.

7. Conclusion

Across 90 controlled runs, crop host supervision with equal loss weight did not provide a dependable advantage over a flat Swin Small classifier. All primary outcomes were equivalent on NPD-Rice-COCO-43. Accuracy and CHER were equivalent on MultiCrop-88, while the remaining outcomes were unresolved at the ± 0.25 point margin. Neither agreement loss improved every dataset and metric. Predicted-host masking removed head conflicts but had little effect on accuracy. In contrast, the labeled-host oracle improved

accuracy on the benchmark with 215 classes by about 2.2 points. Independent host information can be useful, but agreement produced from the model’s own host prediction is not an independent check of correctness. Temperature scaling gave the most consistent improvement in probability quality. Under the tested conditions, the host head is better viewed as an audit channel than as a reliable route to higher condition accuracy.

Data and code availability

Code for training, evaluation, statistical analysis, and figure generation is available in the GitHub repository. The repository contains the exact split manifests, ordered taxonomies, configuration records, metrics for each seed, figure scripts, artifact hashes, and the Agri-Foundation-145k placeholder-class sensitivity output described in the Supplementary Material. The image datasets are not redistributed because each source has its own license and terms. Logits and checkpoints should be deposited separately where the source terms and storage limits allow.

CRedit authorship contribution statement

Muhammad Omair Khan and Muhammad Ahmad contributed equally to this work. Muhammad Omair Khan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft. Muhammad Ahmad: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft. MD Ariful Islam Mozumder: Methodology, Validation, Resources, Writing - review and editing. Hee-Cheol Kim: Supervision, Project administration, Resources, Writing - review and editing, Corresponding author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the authors used a large language model (LLM) for language editing, manuscript organization, reference checking, LaTeX preparation, code review, and figure preparation. The authors reviewed and edited the resulting material, checked the references and numerical statements against the source files and stored results, and take full responsibility for the content of the article.

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